

Roll No.....

Mid Semester Examination September 2025

B.Tech(ME)
Microprocessor and Microcontroller

Semester: IIIrd
Paper Code: TME-311

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Explain difference between Microprocessor and Micro controller.	
	OR	CO1
(b)	Explain the program written below ADD A, 45H SUBB A, #55H. MOV DPTR, #2000H MUL AB	CO2
Q2	(10 marks)	
(a)	Explain difference between RISC based CPU and CISC based CPU .	CO1
	OR	CO2
(b)	Discuss immediate addressing mode and register addressing mode along with example .	
Q3	(10 marks)	
(a)	Discuss 8051 microcontroller architecture along with block diagram	CO1
	OR	CO2
(b)	Discuss Direct addressing and Indirect addressing mode along with example .	
Q4	(10 marks)	
(a)	Draw the pin diagram 8051 microcontroller and explain function of pins.	CO1
	OR	CO2
(b)	Explain relative addressing and Absolute addressing mode along with examples	
Q5	(10 marks)	
(a)	Explain components of CPU of computer.	CO1
	OR	CO2
(b)	What do you understand Long addressing and Indexed addressing along with examples.	

